

YASH KARTHIYA

Full-Stack Developer | B.Tech Student

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Full-Stack Engineer (Backend-Leaning) with project experience in Java/Spring Boot, React, and real-time applications. Focused on performance optimization, scalable system design, and building intuitive end-to-end product features.

TECHNICAL EXPERTISE

Languages	Java, Python, JavaScript (ES6+), SQL
Frontend	React, Tailwind CSS, Context API, React Router, Axios
Backend	Spring Boot, Express, FastAPI, JWT Auth, Socket.io, WebRTC, REST APIs
Tools & Infra	Docker, AWS (EC2, S3), Redis, PostgreSQL, MongoDB, Git, Linux

EDUCATION

Bachelor of Technology in Computer Engineering

Dharmsinh Desai University | CGPA: 8.29 / 10.0

2023 – 2027
Nadiad, Gujarat

KEY PROJECTS

SkillSwap: Real-Time Peer Learning Platform

Jul 2025 – Nov 2025

Node.js, React, MongoDB, Socket.io, WebRTC

[View on GitHub](#)

- Built a real-time peer-learning platform using Node.js, React, Socket.io, and WebRTC; supported 50+ concurrent sessions with sub-100ms connection setup.
- Managed complex call states and real-time user availability in React, utilizing the Context API to ensure cross-component synchronization during active sessions.
- Implemented JWT-based authentication and automated session cleanup, reducing stale connection overhead by 25% while maintaining a seamless user experience.

Full-Stack Food Label AI Scanner (DUHacks 5.0)

2026

FastAPI, React, Tesseract, OpenAI

[View on GitHub](#)

- Designed an intuitive React interface for real-time health-score visualization, tested and validated across 200+ product scans.
- Engineered a FastAPI-based OCR-to-NLP pipeline to convert unstructured packaging text into structured JSON data for real-time health analysis.
- Deployed the full-stack system on AWS for live field testing and demonstration during the hackathon, ensuring mobile responsiveness.

Scalable Shop Inventory Management System

Dec 2024 – Mar 2025

Spring Boot, PostgreSQL, Redis, Docker

[View on GitHub](#)

- Eliminated database bottlenecks via Redis caching for hot SKUs, achieving sub-50ms latency in high-load benchmark tests.
- Implemented asynchronous state updates with Spring Events to decouple inventory logging from API threads, improving system stability under high-traffic scenarios.

RAG-Based Academic Query Engine

Dec 2025 – Mar 2026

FastAPI, Python, ChromaDB, LangChain

[View on GitHub](#)

- Built an AI-powered query engine that improved response relevance for academic documents by 30% using optimized vector indexing and LangChain retrieval.
- Achieved sub-500ms response times for large PDF datasets using recursive character splitting and personalized context-aware retrieval.

ADDITIONAL INFORMATION

- Deployment:** Deployed multiple project demos on AWS for live testing and peer feedback.
- Hackathon:** Built and demoed a full-stack food-label analysis system at DUHacks 5.0.
- Languages:** English (Professional), Hindi (Native), Gujarati (Native)